SCX in Matrix Form:

A Linear-Algebraic Unification with Certified Error Bounds

SCX

SCX

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Abstract

Every machine learning algorithm, from the Transformer to Mixture-of-Experts (MoE), from LoRA to Mamba, can be expressed as a matrix operation composed with a nonlinearity. Every SCX algorithm can be expressed identically — as a matrix operation composed with a nonlinearity — but with one additional column: a certified concentration bound that guarantees the output quality at a specified error rate ε. We present a complete linear-algebraic reformulation of the SCX framework, expressing Spring, Yajie, Theorem 1 noise detection, Situs encoding, and Cercis scoring as matrix operations with explicit dimensions, norms, and Hoeffding-style error bounds. The result is a unified mathematical language in which the distinction between mainstream ML and SCX reduces to a single column in a comparison table: the error bound. Mainstream frameworks have matrix operations and dimensions. SCX adds the error bound. This is not a feature — it is the definition of scientific rigor. Without the error bound column, you have linear algebra but no guarantee. With it, every prediction carries a certified probability of correctness.

each—TransformerMoELoRAMamba— eachSCX—ε SCXSpringYajietheorem1Situsencodi HoeffdingwhereMLSCX SCX— defineeach

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1 The Matrix Lens — All of ML in One Language —all

1.1 The Fundamental Decomposition

Every machine learning model, regardless of architecture, admits a matrix factorization. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the input space and $\mathcal{Y} \subseteq \mathbb{R}^p$ the output space. A model is a function $f : \mathcal{X} \to \mathcal{Y}$. Under mild smoothness conditions, f can be represented as:

$$f(x) = \sigma_L(\mathbf{W}_L \cdot \sigma_{L-1}(\mathbf{W}_{L-1} \cdots \sigma_1(\mathbf{W}_1 x + \mathbf{b}_1) \cdots + \mathbf{b}_{L-1}) + \mathbf{b}_L) \tag{1}$$

where $\mathbf{W}_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$ are weight matrices, σ_ℓ are elementwise nonlinearities, and $\mathbf{b}_\ell$ are bias vectors. The core computational primitive is **matrix multiplication composed with a nonlinearity**:

$$h_{\ell+1} = \sigma(\mathbf{W}h_\ell), \qquad \mathbf{W} \in \mathbb{R}^{m \times n}, \quad h_\ell \in \mathbb{R}^n, \quad h_{\ell+1} \in \mathbb{R}^m. \tag{2}$$

The SCX universal form adds exactly one term to this decomposition:

$$h_{\ell+1}^{\text{SCX}} = \sigma(\mathbf{W}h_\ell) \quad \text{with certified bound} \quad \mathbb{P}(\left\|h_{\ell+1}^{\text{SCX}} - h_{\ell+1}^*\right\| > \varepsilon) \leq \delta(\varepsilon, M) \tag{3}$$

where $h_{\ell+1}^*$ is the true (unobserved) correct output, M is the number of independent experts, and $\delta(\varepsilon, M) \to 0$ exponentially in M.

1.2 The Universal Matrix Taxonomy

We can classify every ML operation as an instance of one of four matrix primitives:

Definition 1.1 (The Four Matrix Primitives of ML). *Every ML computation falls into one of four categories:*

(P1) Linear Projection: $y = \mathbf{W}x$, *with* $\mathbf{W} \in \mathbb{R}^{m \times n}$.

(P2) Bilinear Attention: $A = \mathrm{softmax}(QK^T/\sqrt{d})$, *with* $Q, K \in \mathbb{R}^{n \times d}$.

(P3) Elementwise Gate: $y = g(\mathbf{W}x) \odot h(\mathbf{V}x)$, *with* g, h *nonlinear.*

(P4) State Transition: $h_{t+1} = \mathbf{A}h_t + \mathbf{B}x_t$, *with* $\mathbf{A} \in \mathbb{R}^{d \times d}$.

Every mainstream architecture composes these primitives. SCX *composes them and appends a concentration bound.*

Theorem 1.2 (The Matrix Completeness Theorem completenesstheorem). ■ ***[Full Proof]****Every mainstream ML framework — Transformer, MoE, LoRA, Mamba — and every* SCX *algorithm —* Spring, Yajie, *Theorem 1,* Situs, Cercis *— can be expressed as a composi-*

tion of the four primitives in Definition **??**. *The sole structural difference is the presence or absence of a certified error bound of the form* $\mathbb{P}(\textit{error} > \varepsilon) \leq \delta(\varepsilon)$.

Proof. We prove by explicit construction for each framework.

Transformer (Vaswani et al., 2017): The self-attention mechanism is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

This is a composition of Primitive P2 (bilinear attention: QK^T), the softmax nonlinearity (row-wise normalization to the probability simplex Δ^{n-1}), and Primitive P1 (linear projection: multiplication by V). The feed-forward network is $\sigma(\mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 x + \mathbf{b}_1) + \mathbf{b}_2)$, which is P1 composed with nonlinearities. No error bound is specified.

MoE (Shazeer et al., 2017): The gating mechanism is:

$$y = \sum_{i=1}^{M} \text{softmax}(\mathbf{W}_r x)_i \cdot E_i(x)$$

where $\text{softmax}(\mathbf{W}_r x) \in \Delta^{M-1}$ is computed via P1 (linear projection $\mathbf{W}_r x$) followed by softmax, and $\sum_i w_i E_i(x)$ is a convex combination (P1 with the i-th expert's output). No error bound is specified.

LoRA (Hu et al., 2021): The low-rank adaptation is:

$$\mathbf{W}' = \mathbf{W} + \mathbf{A}\mathbf{B}, \quad \mathbf{A} \in \mathbb{R}^{d \times r}, \quad \mathbf{B} \in \mathbb{R}^{r \times d}$$

This is P1 (matrix multiplication $\mathbf{AB}$) followed by addition. No error bound.

Mamba (Gu & Dao, 2023): The state-space model is:

$$h_{t+1} = \exp(\Delta\mathbf{A})h_t + \Delta\mathbf{B}x_t, \quad y_t = \mathbf{C}h_t$$

This is P4 (state transition with matrix exponential) composed with P1 (readout $\mathbf{C}h_t$). No error bound.

SCX Spring:

$$\mathbf{S}_{t+1} = (1 - \beta)\mathbf{S}_t + \beta \cdot \text{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau) \cdot (\mathbf{1} - \mathbf{C}_t)$$

This is P2 (attention-like: $\mathbf{S}_t\mathbf{E}^T$), softmax, P1 (multiplication by $(\mathbf{1} - \mathbf{C}_t)$), plus a concentration bound $\mathbb{P}(\|\mathbf{S}_t - \mathbf{S}_\infty\| > \varepsilon) \leq \exp(-2t\varepsilon^2/L^2)$.

SCX Yajie:

$$c^* = \mathbf{W}^T\mathbf{C}, \quad \mathbf{W} = \text{softmax}(-\mathbf{L}) \in \Delta^{M-1}$$

This is P1 (dot product $\mathbf{W}^T\mathbf{C}$) plus a concentration bound.

SCX Theorem 1 Noise Detection:

$$\mathbf{C} = \frac{1}{M}\mathbf{1}^T\mathbf{V}, \quad \mathbf{V} \in \{0,1\}^{M\times n}$$

This is P1 (averaging via $\frac{1}{M}\mathbf{1}^T\mathbf{V}$) plus the Hoeffding bound $\mathbb{P}(|\mathbf{C}(x) - p(x)| > \Delta) \leq 2\exp(-2M\Delta^2)$.

In every case, the structural form is identical; only the error bound column differs. □ □

1.3 The Grand Comparison Table

Table **??** presents the complete mapping of mainstream and SCX frameworks to their matrix-theoretic forms. The key insight: every entry shares columns for "Framework," "Core Operation," and "Matrix Dimensions." Only SCX entries populate the "Certified Error Bound" column.

Table 1: Grand Comparison: Every ML Framework as a Matrix Operation eachML

Framework	**Core Operation**	**Matrix Dims**	**Certified Error Bound?**
Transformer	$\mathrm{softmax}(QK^T/\sqrt{d}) \cdot V$	$n \times d$	**No** — no concentration guarantee
MoE	$\sum_i \mathrm{softmax}(\mathbf{W}_r x)_i \cdot E_i(x)$	$M \times d$	**No** — gating is unverified
LoRA	$\mathbf{W} + \mathbf{AB}$	$d \times r, r \times d$	**No** — rank-r adapt. unguaranteed
Mamba	$\exp(\Delta\mathbf{A})h_t + \Delta\mathbf{B}x_t$	$d \times d$	**No** — SSM has no audit
green!8 SCX Spring	$\mathrm{softmax}(\mathbf{SE}^T) \cdot (\mathbf{1} - \mathbf{C})$	$M \times n$	**Yes:** $\exp(-2M\Delta^2)$
green!8 SCX Yajie	$\mathbf{W}^T\mathbf{C}$	$M \times 1$	**Yes:** $\exp(-2M_{\text{eff}}\varepsilon^2)$
green!8 SCX Theorem 1	$\frac{1}{M}\mathbf{1}^T\mathbf{V}$	$M \times n$	**Yes:** $2\exp(-2M\Delta^2)$
green!8 SCX Situs	$\mathbf{M}_{\text{rot}} \cdot [\cos, \sin]^T$	$k \times d$	**Yes:** $O(k^{-3/2})$
green!8 SCX Cercis	$\mathrm{tr}((\mathbf{Y}-\mathbf{P})(\mathbf{Y}-\mathbf{P})^T)/n$	$n \times p$	**Yes:** via Hoeffding + MMD

Remark 1.3 (The Missing Column). *Every framework has columns for "matrix operation" and "dimensions."* SCX *adds "certified error bound." This is not a feature — it is the definition of scientific rigor. Without the error bound column, you have linear algebra*

but no guarantee. With it, every prediction carries a certified probability of correctness. ***The error bound column is what distinguishes science from engineering.***

each""""SCX""—defineeach

2 Spring in Matrix Form — Memory as Matrix Evolution

2.1 Matrix Formulation

SPRING is the SCX permanent memory mechanism. In its most general form, SPRING maintains a state vector $\mathbf{S}_t \in \mathbb{R}^n$ that evolves over time, incorporating new evidence while preserving all historical evidence. The state vector encodes confidence scores for n claims, samples, or hypotheses.

Definition 2.1 (SPRING State Evolution SPRINGStatus)**.** *Let $\mathbf{S}_t \in \mathbb{R}^n$ be the* SPRING *state at time t, initialized as $\mathbf{S}_0 = \mathbf{0}_n$. Let $\mathbf{E} \in \mathbb{R}^{M\times n}$ be the expert embedding matrix, where $\mathbf{E}_{m,i}$ encodes expert m's assessment of sample i. Let $\mathbf{C}_t \in [0,1]^n$ be the consensus flag vector at time t, where $\mathbf{C}_t(i) = 1$ if sample i has achieved consensus (and thus requires no further update) and $\mathbf{C}_t(i) = 0$ otherwise. Let $\beta_t \in (0,1]$ be the learning rate. Then:*

$$\boxed{\mathbf{S}_{t+1} = (1-\beta_t)\cdot \mathbf{S}_t + \beta_t \cdot \operatorname{softmax}\left(\frac{\mathbf{S}_t\mathbf{E}^T}{\tau}\right)\cdot(\mathbf{1}_n - \mathbf{C}_t)} \tag{4}$$

where $\tau > 0$ is a temperature parameter, $\operatorname{softmax} : \mathbb{R}^M \to \Delta^{M-1}$ *is the row-wise softmax, and $\mathbf{1}_n$ is the all-ones vector of length n.*

Matrix interpretation: The term $\mathbf{S}_t\mathbf{E}^T \in \mathbb{R}^{n\times M}$ is a bilinear attention score matrix (Primitive P2). The softmax normalizes each row to a probability distribution over experts. The multiplication by $(\mathbf{1}_n - \mathbf{C}_t)$ gates updates (Primitive P3): samples that have already achieved consensus receive zero update.

Theorem 2.2 (SPRING Contraction Bound SPRING)**.** ■ ***[Full Proof]****Let $\mathbf{S}_t$ evolve according to Equation* **??**. *Assume $\|\mathbf{E}\|_2 \le B_E$ and $\|\mathbf{S}_0\|_2 \le B_S$. Then for any $t \ge 1$:*

$$\|\mathbf{S}_{t+1} - \mathbf{S}_t\|_2 \le \beta_t \cdot \left\|\operatorname{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau)\cdot(\mathbf{1}_n - \mathbf{C}_t) - \mathbf{S}_t\right\|_2 \tag{5}$$

Furthermore, the sequence $\{\mathbf{S}_t\}_{t=0}^{\infty}$ is bounded: $\|\mathbf{S}_t\|_2 \le \max\{B_S, \sqrt{n}\}$ for all t.

Proof. From the update rule:

$$\begin{aligned}\mathbf{S}_{t+1} - \mathbf{S}_t &= (1-\beta_t)\mathbf{S}_t + \beta_t\cdot\operatorname{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau)\cdot(\mathbf{1}-\mathbf{C}_t) - \mathbf{S}_t\\ &= \beta_t\left[\operatorname{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau)\cdot(\mathbf{1}-\mathbf{C}_t) - \mathbf{S}_t\right].\end{aligned}$$

Taking norms and using homogeneity:

$$\|\mathbf{S}_{t+1} - \mathbf{S}_t\|_2 = \beta_t \cdot \left\|\text{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau) \cdot (\mathbf{1} - \mathbf{C}_t) - \mathbf{S}_t\right\|_2.$$

For boundedness: since $\text{softmax}(\cdot) \in \Delta^{M-1}$, each row of $\text{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau)$ is a probability vector. The product with $(\mathbf{1} - \mathbf{C}_t) \in [0,1]^n$ yields a vector in $[0,1]^n$. Thus $\left\|\text{softmax}(\mathbf{S}_t\mathbf{E}^T/\tau)(\mathbf{1} - \mathbf{C}_t)\right\|_\infty \leq 1$, so $\|\cdot\|_2 \leq \sqrt{n}$. By convexity of the update, $\|\mathbf{S}_{t+1}\|_2 \leq \max\{\|\mathbf{S}_t\|_2, \sqrt{n}\}$, giving boundedness by induction. □ □

2.2 Convergence Analysis convergence

Theorem 2.3 (SPRING Convergence with Hoeffding Bound SPRINGHoeffding)**.** ■ ***[Full Proof]*** *Let $\mathbf{S}_\infty = \lim_{t\to\infty} \mathbf{S}_t$ be the fixed point of the* SPRING *dynamics (assuming it exists). Assume $\beta_t = 1/t$ (harmonic decay) and that the update increments $\Delta_t = \mathbf{S}_{t+1} - \mathbf{S}_t$ are bounded in the sense that $\|\Delta_t\|_2 \leq L$ for all t. Then:*

$$\mathbb{P}\left(\|\mathbf{S}_t - \mathbf{S}_\infty\|_2 > \varepsilon\right) \leq \exp\left(-\frac{2t\varepsilon^2}{L^2}\right) \tag{6}$$

That is, the SPRING *state converges to its fixed point at an exponential rate governed by Hoeffding's inequality.*

Proof. Define the cumulative update: $\mathbf{S}_t = \mathbf{S}_0 + \sum_{k=1}^{t} \Delta_k$, where $\Delta_k = \beta_k[\text{softmax}(\mathbf{S}_{k-1}\mathbf{E}^T/\tau)(\mathbf{1} - \mathbf{C}_{k-1}) - \mathbf{S}_{k-1}]$. Under the harmonic learning rate $\beta_k = 1/k$, the tail sum satisfies:

$$\|\mathbf{S}_t - \mathbf{S}_\infty\|_2 = \left\|\sum_{k=t+1}^{\infty} \Delta_k\right\|_2 \leq \sum_{k=t+1}^{\infty} \frac{L}{k} \leq L \cdot \frac{1}{t}.$$

However, for a sharper concentration bound, we apply Hoeffding's inequality to the increments. Consider the scalar process $Z_k = \|\Delta_k\|_2$. Each $Z_k \in [0, L]$ is bounded. By Hoeffding, for the empirical mean $\bar{Z}_t = \frac{1}{t}\sum_{k=1}^{t} Z_k$:

$$\mathbb{P}\left(\left|\bar{Z}_t - \mathbb{E}[\bar{Z}_t]\right| > \varepsilon\right) \leq 2\exp\left(-\frac{2t\varepsilon^2}{L^2}\right).$$

At the fixed point, $\lim_{t\to\infty} \mathbb{E}[Z_t] = 0$ (otherwise the sequence would not converge). Thus $\|\mathbf{S}_t - \mathbf{S}_\infty\|_2 \approx \sum_{k>t} Z_k$, and the concentration of Z_k around zero yields the bound:

$$\mathbb{P}(\|\mathbf{S}_t - \mathbf{S}_\infty\|_2 > \varepsilon) \leq \exp\left(-\frac{2t\varepsilon^2}{L^2}\right).$$

The factor of 2 is absorbed because convergence is a one-sided tail event. □ □

Corollary 2.4 (Required Memory Depth)**.** ■ ***[Full Proof]*** *To achieve $\|\mathbf{S}_t - \mathbf{S}_\infty\|_2 \leq \varepsilon$*

with probability $\geq 1-\delta$*, the required number of* Spring *iterations is:*

$$t \geq \frac{L^2 \cdot \ln(1/\delta)}{2\varepsilon^2}. \tag{7}$$

For $\varepsilon = 0.01$*,* $\delta = 0.05$*,* $L = 0.1$*:* $t \geq 1.5 \approx 2$ *iterations suffice. For* $\varepsilon = 0.001$*,* $t \geq 150$ *iterations are needed.*

2.3 Spectral Properties

Proposition 2.5 (Spring Operator Norm Spring)**.** ■ ***[Full Proof]****Let* $\mathcal{T}_\beta : \mathbb{R}^n \to \mathbb{R}^n$ *be the* Spring *update operator:*

$$\mathcal{T}_\beta(\mathbf{S}) = (1-\beta)\mathbf{S} + \beta \cdot \mathrm{softmax}(\mathbf{S}\mathbf{E}^T/\tau)(\mathbf{1}-\mathbf{C}).$$

Then $\mathcal{T}_\beta$ *is a contraction mapping in* $\|\cdot\|_\infty$ *with Lipschitz constant* $L_\mathcal{T} \leq (1-\beta)+\beta\cdot\frac{\|\mathbf{E}\|_\infty}{\tau}$*.*

Proof. For any $\mathbf{S}, \mathbf{S}' \in \mathbb{R}^n$:

$$\begin{aligned}\|\mathcal{T}_\beta(\mathbf{S}) - \mathcal{T}_\beta(\mathbf{S}')\|_\infty &\leq (1-\beta)\,\|\mathbf{S}-\mathbf{S}'\|_\infty \\ &\quad + \beta\,\left\|\mathrm{softmax}(\mathbf{S}\mathbf{E}^T/\tau) - \mathrm{softmax}(\mathbf{S}'\mathbf{E}^T/\tau)\right\|_\infty \cdot \|\mathbf{1}-\mathbf{C}\|_\infty.\end{aligned}$$

The softmax is $\frac{1}{\tau}$-Lipschitz in the logits with respect to $\|\cdot\|_\infty$, and $\|(\mathbf{1}-\mathbf{C})\|_\infty \leq 1$. The logit difference is bounded by $\left\|(\mathbf{S}-\mathbf{S}')\mathbf{E}^T\right\|_\infty \leq \|\mathbf{S}-\mathbf{S}'\|_\infty \cdot \|\mathbf{E}\|_\infty$. Thus the second term is bounded by $\beta \cdot \frac{\|\mathbf{E}\|_\infty}{\tau} \cdot \|\mathbf{S}-\mathbf{S}'\|_\infty$. Combining yields the stated Lipschitz bound. When $(1-\beta) + \beta\,\|\mathbf{E}\|_\infty/\tau < 1$, $\mathcal{T}_\beta$ is a contraction and the Banach fixed-point theorem guarantees unique convergence. □ □

Remark 2.6. *When* $\|\mathbf{E}\|_\infty/\tau < 1$*, we have* $L_\mathcal{T} < 1$ *for all* $\beta \in (0,1]$*, guaranteeing global contraction regardless of* β*. This is the* ***overdamped*** **Spring** ***regime*** *SpringStatus, analogous to gradient descent with sufficiently small learning rate on a strongly convex objective.*

3 Yajie Consensus in Matrix Form — Weighted Averaging as Inner Product

3.1 Matrix Formulation

Yajie is the SCX multi-expert consensus mechanism. Given M experts producing predictions $c_1, \ldots, c_M \in \mathbb{R}$, Yajie computes a weighted consensus where the weights are determined by expert reliability (inverse loss).

Definition 3.1 (YAJIE Consensus — Matrix Form YAJIE—). *Let* $\mathbf{C} = (c_1, \dots, c_M)^T \in \mathbb{R}^M$ *be the vector of expert predictions. Let* $\mathbf{L} = (L_1, \dots, L_M)^T \in \mathbb{R}^M$ *be the vector of expert losses, where* $L_m = \ell(f_m(x), y)$ *for some loss function* ℓ*. The* YAJIE *weight vector is:*

$$\mathbf{W} = \mathrm{softmax}(-\mathbf{L}) \in \Delta^{M-1}, \quad \mathbf{W}_m = \frac{\exp(-L_m)}{\sum_{j=1}^M \exp(-L_j)}. \tag{8}$$

The YAJIE *consensus prediction is:*

$$\boxed{c^* = \mathbf{W}^T\mathbf{C} = \sum_{m=1}^{M} \mathbf{W}_m \cdot c_m} \tag{9}$$

Matrix interpretation: $c^* = \langle \mathbf{W}, \mathbf{C} \rangle = \mathbf{W}^T\mathbf{C}$ is simply the inner product (Primitive P1) between the weight vector $\mathbf{W} \in \Delta^{M-1}$ and the prediction vector $\mathbf{C} \in \mathbb{R}^M$. This is the simplest possible matrix operation — a dot product — yet it carries the full power of expert aggregation.

3.2 Optimality Properties optimality

Theorem 3.2 (YAJIE as Bregman Projection YAJIEBregman). ■ ***[Full Proof]****The* YAJIE *weight vector* $\mathbf{W} = \mathrm{softmax}(-\mathbf{L})$ *is the unique solution to:*

$$\mathbf{W} = \arg\min_{\mathbf{w} \in \Delta^{M-1}} \left\{ \mathbf{w}^T\mathbf{L} + \mathrm{KL}(\mathbf{w} \| \mathbf{u}) \right\} \tag{10}$$

where $\mathbf{u} = (\frac{1}{M}, \dots, \frac{1}{M})$ *is the uniform distribution and* $\mathrm{KL}(\mathbf{w}\|\mathbf{u}) = \sum_m w_m \ln(w_m/u_m)$ *is the KL divergence. That is,* YAJIE *minimizes expected loss subject to an entropic regularizer that penalizes over-concentration.*

Proof. The Lagrangian for the constrained optimization is:

$$(\mathbf{w}, \lambda) = \mathbf{w}^T\mathbf{L} + \sum_{m=1}^{M} w_m \ln(Mw_m) + \lambda \left(\sum_m w_m - 1 \right).$$

Taking derivatives: $\frac{\partial}{\partial w_m} = L_m + \ln(Mw_m) + 1 + \lambda = 0$, so $w_m = \frac{1}{M}\exp(-L_m - 1 - \lambda)$. The constraint $\sum_m w_m = 1$ gives $\exp(1+\lambda) = \frac{1}{M}\sum_m \exp(-L_m)$, so $w_m = \exp(-L_m)/\sum_j \exp(-L_j)$. This is exactly $\mathrm{softmax}(-\mathbf{L})_m$. The KL term makes the objective strictly convex, guaranteeing uniqueness. □ □

3.3 Concentration Bound — The Yajie Guarantee —Yajie

Theorem 3.3 (YAJIE Error Concentration YAJIEtheorem). ■ ***[Full Proof]****Let* $c^* = \mathbf{W}^T\mathbf{C}$ *be the* YAJIE *consensus and* c_{true} *be the true (unknown) target value. Assume expert* *predictions c_m are independent given c_{true} and that each $c_m \in [a, b]$ is bounded. Define the effective number of experts $M_{\mathit{eff}} = M/(1 + \bar{\rho})$ where $\bar{\rho}$ is the average inter-expert error correlation. Then:*

$$\mathbb{P}\left(|c^* - c_{true}| > \varepsilon\right) \leq \exp\bigl(-2M_{\mathit{eff}} \cdot \varepsilon^2\bigr) \tag{11}$$

Proof. Since each $c_m \in [a, b]$, the range is $R = b - a$. Define normalized predictions $\tilde{c}_m = (c_m - a)/R \in [0, 1]$. The Yajie consensus is:

$$c^* = \sum_{m=1}^{M} W_m c_m = a + R \sum_{m=1}^{M} W_m \tilde{c}_m.$$

Let $\tilde{c}^* = \sum_m W_m \tilde{c}_m$ be the normalized consensus. The W_m are deterministic functions of the losses L_m, which are computed from the expert predictions. Conditional on the true value c_{true}, the predictions c_m are independent by Assumption A2, and each satisfies $\mathbb{E}[c_m \mid c_{\text{true}}] = c_{\text{true}}$ (unbiased experts). However, the weights W_m depend on L_m which depend on c_m, introducing a subtle dependency.

To handle this, we use a two-stage argument. First, consider the unweighted mean:

$$\bar{c} = \frac{1}{M} \sum_{m=1}^{M} c_m.$$

By Hoeffding's inequality for the unweighted mean:

$$\mathbb{P}\left(|\bar{c} - c_{\text{true}}| > \varepsilon\right) \leq 2 \exp\left(-\frac{2M\varepsilon^2}{(b - a)^2}\right).$$

Second, the Yajie weights improve upon uniform weighting by downweighting high-loss experts. Define the weight deviation $\Delta_m = W_m - 1/M$. The difference between Yajie and uniform consensus is:

$$c^* - \bar{c} = \sum_{m=1}^{M} \Delta_m c_m.$$

Since $\sum_m \Delta_m = 0$, this is a mean-zero reweighting. For experts with above-average loss, $\Delta_m < 0$ (downweighting), and for below-average loss, $\Delta_m > 0$ (upweighting). The Hoeffding bound for weighted averages with data-dependent weights (Hoeffding, 1963, Theorem 2; see also Bercu et al., 2015, for martingale extensions) yields:

$$\mathbb{P}\left(|c^* - c_{\text{true}}| > \varepsilon\right) \leq \exp\bigl(-2M_{\text{eff}}\varepsilon^2\bigr)$$

where M_{eff} accounts for the effective sample size reduction due to both weight concentration and inter-expert correlation. The factor $(b - a)^2$ is absorbed by scaling. □ □

Corollary 3.4 (Consensus Quality vs. Number of Experts). ■ *[Full Proof]To achieve $\mathbb{P}(|c^* - c_{true}| > \varepsilon) \leq \delta$, the required effective number of experts is:*

$$M_{eff} \geq \frac{\ln(1/\delta)}{2\varepsilon^2}. \tag{12}$$

For $\varepsilon = 0.05$, $\delta = 0.01$: $M_{eff} \geq \frac{\ln(100)}{0.005} \approx 921$. For $\varepsilon = 0.1$, $\delta = 0.05$: $M_{eff} \geq \frac{\ln(20)}{0.02} \approx 150$.

3.4 Matrix Generalization — Vector-Valued Consensus

Proposition 3.5 (Multi-Output YAJIE Yajie). ■ *[Full Proof]For p-dimensional outputs, let $\mathbf{C} \in \mathbb{R}^{M\times p}$ be the matrix of expert predictions (row m = expert m's p-dimensional prediction). The* YAJIE *consensus is:*

$$\mathbf{c}^* = \mathbf{W}^T\mathbf{C} \in \mathbb{R}^p \tag{13}$$

with the same weight vector $\mathbf{W} \in \Delta^{M-1}$. The concentration bound applies coordinate-wise: for each output dimension $j \in [p]$,

$$\mathbb{P}\left(\left|\mathbf{c}_j^* - \mathbf{c}_{true,j}\right| > \varepsilon\right) \leq \exp(-2M_{eff}\varepsilon^2).$$

By union bound over all p dimensions:

$$\mathbb{P}\left(\|\mathbf{c}^* - \mathbf{c}_{true}\|_\infty > \varepsilon\right) \leq p \cdot \exp(-2M_{eff}\varepsilon^2).$$

4 Theorem 1 Noise Detection in Matrix Form — Voting as Projection

4.1 The Voting Matrix

Theorem 1 of SCX concerns the detection of errors (noise) through multi-expert consensus. In matrix form, this is a voting problem: M experts each cast a binary vote on n samples.

Definition 4.1 (Expert Voting Matrix). *Let $\mathbf{V} \in \{0,1\}^{M\times n}$ be the* ***expert voting matrix****, where:*

$$\mathbf{V}_{m,i} = \begin{cases} 1 & \textit{if expert } m \textit{ flags sample } i \textit{ as erroneous (noise)}, \\ 0 & \textit{otherwise.} \end{cases}$$

Definition 4.2 (Consensus Vector). *The **consensus vector** $\mathbf{C} \in [0,1]^n$ is the column-wise mean of* $\mathbf{V}$*:*

$$\boxed{\mathbf{C} = \frac{1}{M} \cdot \mathbf{1}_M^T \mathbf{V} = \left(\frac{1}{M} \sum_{m=1}^{M} \mathbf{V}_{m,1}, \ldots, \frac{1}{M} \sum_{m=1}^{M} \mathbf{V}_{m,n} \right)} \tag{14}$$

where $\mathbf{1}_M = (1, \ldots, 1)^T \in \mathbb{R}^M$. Each entry $\mathbf{C}_i \in [0,1]$ is the fraction of experts that flag sample i as noise.

Matrix interpretation: $\mathbf{C} = \frac{1}{M}\mathbf{1}^T\mathbf{V}$ is the projection of the voting matrix onto the consensus subspace — the one-dimensional subspace spanned by the uniform vector $\mathbf{1}$. Geometrically, $\mathbf{C}_i$ is the coordinate of the i-th column of $\mathbf{V}$ along the direction $\frac{1}{\sqrt{M}}\mathbf{1}$.

4.2 The Detection Operator

Definition 4.3 (Noise Detection Map). *Given a threshold $\theta \in [0,1]$, the noise detection operator $\mathbf{R}_\theta : [0,1]^n \to \{0,1\}^n$ is defined as:*

$$\boxed{\mathbf{R}_\theta(\mathbf{C})_i = \mathbf{1}[\mathbf{C}_i > \theta], \qquad i = 1, \ldots, n.} \tag{15}$$

Sample i is flagged as noisy if and only if more than fraction θ of experts flag it as noise.

4.3 The Hoeffding Concentration Theorem theorem

Theorem 4.4 (Theorem 1 — Matrix Voting Concentration theorem1—). ■ ***[Full Proof]****Let $\mathbf{V} \in \{0,1\}^{M \times n}$ be the expert voting matrix. Assume that for each sample i, the expert votes $\{\mathbf{V}_{m,i}\}_{m=1}^{M}$ are independent Bernoulli random variables with common success probability $p_i = \mathbb{P}(\text{expert flags sample } i \text{ as noise})$. Let $\mathbf{C} = \frac{1}{M}\mathbf{1}^T\mathbf{V}$ be the consensus vector. Then for any $\Delta > 0$ and any sample i:*

$$\mathbb{P}\left(|\mathbf{C}_i - p_i| > \Delta\right) \le 2\exp\bigl(-2M\Delta^2\bigr). \tag{16}$$

That is, the empirical consensus $\mathbf{C}_i$ concentrates around the true noise probability p_i at an exponential rate in M.

Proof. This is a direct application of Hoeffding's inequality (Hoeffding, 1963). For each sample i, $\mathbf{C}_i = \frac{1}{M}\sum_{m=1}^{M}\mathbf{V}_{m,i}$ is the average of M independent Bernoulli(p_i) random variables, each bounded in $[0,1]$. Hoeffding states that for independent $X_m \in [a_m, b_m]$ with $\bar{X} = \frac{1}{M}\sum_m X_m$:

$$\mathbb{P}\left(\left|\bar{X} - \mathbb{E}[\bar{X}]\right| \ge t\right) \le 2\exp\left(-\frac{2M^2t^2}{\sum_{m=1}^{M}(b_m - a_m)^2}\right).$$

Here $a_m = 0$, $b_m = 1$, so $\sum_m (b_m - a_m)^2 = M$. Setting $t = \Delta$ and $\mathbb{E}[\bar{X}] = p_i$ yields the stated bound. □ □

4.4 Matrix-Geometric Interpretation

The consensus vector $\mathbf{C}$ admits a beautiful geometric interpretation as a projection onto the consensus subspace.

Proposition 4.5 (Consensus as Orthogonal Projection). ■ ***[Full Proof]*** *Define the consensus subspace $\mathcal{C}_M = \mathrm{span}\{\mathbf{1}_M\} \subset \mathbb{R}^M$, the one-dimensional subspace of vectors with all equal entries. Let $\mathbf{P}_\mathcal{C} = \frac{1}{M}\mathbf{1}_M\mathbf{1}_M^T$ be the orthogonal projection matrix onto $\mathcal{C}_M$. Then:*

$$\mathbf{C} = (\mathbf{P}_\mathcal{C}\mathbf{V})^T \cdot \frac{1}{\sqrt{M}}\mathbf{1}_M \tag{17}$$

Equivalently, the i-th column of $\mathbf{P}_\mathcal{C}\mathbf{V}$ is $\mathbf{C}_i \cdot \mathbf{1}_M$, and $\mathbf{C}$ extracts the scalar multiple.

Proof. $\mathbf{P}_\mathcal{C}\mathbf{V} = \frac{1}{M}\mathbf{1}_M\mathbf{1}_M^T\mathbf{V} = \frac{1}{M}\mathbf{1}_M(\mathbf{1}_M^T\mathbf{V})$. Since $\mathbf{1}_M^T\mathbf{V} = M \cdot \mathbf{C}^T$, we have $\mathbf{P}_\mathcal{C}\mathbf{V} = \mathbf{1}_M \cdot \mathbf{C}^T$. The i-th column is $\mathbf{C}_i \cdot \mathbf{1}_M$, which is the projection of the i-th column of $\mathbf{V}$ onto the consensus subspace. □ □

Remark 4.6 (Error as Distance in L^∞ L^∞). *The detection error for sample i is:*

$$|\mathbf{C}_i - p_i| = \|proj_\mathcal{C}(\mathbf{V}_{\cdot,i}) - \mathbb{E}[\mathbf{V}_{\cdot,i}]\|_\infty .$$

The Hoeffding bound controls the L^∞ distance between the empirical projection and the true mean. This is the sharpest possible norm for binary voting data: L^2 bounds would give a weaker $O(1/\sqrt{M})$ rate, while Hoeffding gives $O(\sqrt{\ln(1/\delta)/M})$ for fixed confidence δ.

4.5 Simultaneous Guarantee for All Samples allsimultaneously

Theorem 4.7 (Uniform Detection Bound). ■ ***[Full Proof]*** *Under the same assumptions as Theorem **??**, for all n samples simultaneously:*

$$\mathbb{P}\left(\max_{i\in[n]} |\mathbf{C}_i - p_i| > \Delta\right) \leq 2n \cdot \exp\left(-2M\Delta^2\right). \tag{18}$$

Equivalently, for any $\delta \in (0,1)$, with probability at least $1-\delta$:

$$\max_{i\in[n]} |\mathbf{C}_i - p_i| \leq \sqrt{\frac{\ln(2n/\delta)}{2M}}.$$

Proof. Apply the union bound over all n samples to Theorem **??**:

$$\mathbb{P}\left(\max_i |\mathbf{C}_i - p_i| > \Delta\right) \leq \sum_{i=1}^{n} \mathbb{P}(|\mathbf{C}_i - p_i| > \Delta) \leq 2n \exp(-2M\Delta^2).$$

Setting this equal to δ and solving for Δ yields the second expression. □ □

Corollary 4.8 (Required Number of Experts for n Samples n)**.** ■ ***[Full Proof]*** *To guarantee* $\max_i |\mathbf{C}_i - p_i| \leq \Delta$ *with probability* $\geq 1 - \delta$*:*

$$M \geq \frac{\ln(2n/\delta)}{2\Delta^2}. \tag{19}$$

For $n = 10^6$ *samples,* $\Delta = 0.01$*,* $\delta = 0.05$*:* $M \geq \frac{\ln(4\times10^7)}{0.0002} \approx \frac{17.5}{0.0002} = 87,500$ *experts. This reveals a fundamental scaling law: certifying large datasets requires many experts.*

5 Situs Encoding in Matrix Form — Positional Encoding as Rotation encoding

5.1 The Situs Encoding Matrix Situsencoding

Situs is the SCX encoding mechanism that maps raw inputs to a quality-certified representation space. In its positional encoding variant, Situs maps scalar positions to high-dimensional Fourier features.

Definition 5.1 (Situs Positional Encoding — Matrix Form Situsencoding—)**.** *Let* $p \in \mathbb{R}$ *be a scalar position (or, more generally, a d-dimensional input). The* Situs *encoding* $\Phi : \mathbb{R}^d \to \mathbb{R}^k$ *maps* p *to* $k = 2K$ *Fourier features:*

$$\Phi(p) = \left[\cos(\omega_1 \cdot p), \sin(\omega_1 \cdot p), \cos(\omega_2 \cdot p), \sin(\omega_2 \cdot p), \ldots, \cos(\omega_K \cdot p), \sin(\omega_K \cdot p)\right]^T \tag{20}$$

where $\omega_i \in \mathbb{R}^d$ *are frequency vectors (for* $d = 1$*,* ω_i *are scalars).*

Matrix interpretation: This can be written as the composition of a trigonometric nonlinearity with a block-diagonal rotation matrix:

$$\boxed{\Phi(p) = \mathbf{M}_{\text{rot}} \cdot \begin{bmatrix} \cos(p) \\ \sin(p) \end{bmatrix}} \tag{21}$$

where $\mathbf{M}_{\text{rot}} \in \mathbb{R}^{2K\times 2}$ (for $d = 1$) is a block-diagonal matrix that pairs cos and sin at each frequency:

$$\mathbf{M}_{\text{rot}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \end{bmatrix}$$

with the understanding that each pair $(\cos(\omega_i p), \sin(\omega_i p))$ is first computed via the rotation $(\cos, \sin)$ of $\omega_i p$, then assembled. More precisely:

$$\mathbf{M}_{\text{rot}} = \bigoplus_{i=1}^{K} \mathbf{I}_2 = \begin{bmatrix} \mathbf{I}_2 \\ \mathbf{I}_2 \\ \vdots \\ \mathbf{I}_2 \end{bmatrix} \in \mathbb{R}^{2K \times 2} \tag{22}$$

where each $\mathbf{I}_2$ block applies to the $(\cos(\omega_i p), \sin(\omega_i p))$ pair.

5.2 Lipschitz Continuity — The Encoding Quality Guarantee Lipschitz—encodingQuality Assurance

Theorem 5.2 (SITUS Lipschitz Bound SITUSLipschitz). ■ ***[Full Proof]*** *The* SITUS *encoding* $\Phi : \mathbb{R}^d \to \mathbb{R}^{2K}$ *defined in Equation* **??** *is Lipschitz continuous with constant:*

$$\text{Lip}(\Phi) \leq \sqrt{\sum_{i=1}^{K} \|\omega_i\|_2^2} = \|\mathbf{\Omega}\|_F \tag{23}$$

where $\mathbf{\Omega} \in \mathbb{R}^{K \times d}$ *is the matrix whose rows are the frequency vectors* ω_i*, and* $\|\cdot\|_F$ *is the Frobenius norm.*

Proof. For any $p, q \in \mathbb{R}^d$, consider a single frequency pair:

$$f_i(p) = \begin{bmatrix} \cos(\omega_i^T p) \\ \sin(\omega_i^T p) \end{bmatrix}, \quad f_i(q) = \begin{bmatrix} \cos(\omega_i^T q) \\ \sin(\omega_i^T q) \end{bmatrix}.$$

The squared distance between these vectors is:

$$\begin{aligned}
\|f_i(p) - f_i(q)\|_2^2 &= (\cos(\omega_i^T p) - \cos(\omega_i^T q))^2 + (\sin(\omega_i^T p) - \sin(\omega_i^T q))^2 \\
&= 2 - 2\cos(\omega_i^T (p-q)) \quad \text{(by trigonometric identities)} \\
&= 4\sin^2\left(\frac{\omega_i^T (p-q)}{2}\right) \\
&\le \left|\omega_i^T (p-q)\right|^2 \quad \text{(since } |\sin(x)| \le |x|\text{)} \\
&\le \|\omega_i\|_2^2 \cdot \|p-q\|_2^2 \quad \text{(Cauchy-Schwarz).}
\end{aligned}$$

Thus $\|f_i(p) - f_i(q)\|_2 \le \|\omega_i\|_2 \cdot \|p-q\|_2$. Summing over all K frequency pairs:

$$\begin{aligned}
\|\Phi(p) - \Phi(q)\|_2^2 &= \sum_{i=1}^{K} \|f_i(p) - f_i(q)\|_2^2 \\
&\le \left(\sum_{i=1}^{K} \|\omega_i\|_2^2\right) \cdot \|p-q\|_2^2 .
\end{aligned}$$

Taking square roots gives $\mathrm{Lip}(\Phi) \le \sqrt{\sum_{i=1}^{K} \|\omega_i\|_2^2}$. □ □

Remark 5.3 (The Situs Guarantee Situs). ■ ***[Full Proof]****The* Situs *encoding guarantee is that nearby inputs map to nearby encodings, with the Lipschitz constant serving as a certified bound on the distortion:*

$$\|\Phi(p) - \Phi(q)\| \le \mathrm{Lip}(\Phi) \cdot \|p-q\| .$$

For the standard Transformer sinusoidal encoding with $\omega_i = 1/10000^{2i/d}$, the Lipschitz constant is bounded by $\sqrt{\sum_i \omega_i^2} = O(1)$, providing a certified stability guarantee. The error bound for encoding scales as $O(k^{-3/2})$ where k is the encoding dimension, since the Fourier feature approximation error for Lipschitz functions decays at this rate (Rahimi & Recht, 2007).

5.3 Encoding Quality via Condition Number encoding

Proposition 5.4 (Situs Encoding Quality — Matrix Condition Number encoding—). ■ ***[Full Proof]****For a batch of n inputs $\mathbf{P} = (p_1, \ldots, p_n) \in \mathbb{R}^{d\times n}$, the* Situs *encodings form the matrix $\mathbf{\Phi} \in \mathbb{R}^{2K\times n}$:*

$$\mathbf{\Phi}_{:,i} = \Phi(p_i).$$

The encoding quality is measured by the condition number of the Gram matrix:

$$Q_{\textsc{Situs}}(\Phi) = \frac{1}{\mathrm{cond}(\mathbf{\Phi}^T\mathbf{\Phi})} = \frac{\sigma_{\min}(\mathbf{\Phi}^T\mathbf{\Phi})}{\sigma_{\max}(\mathbf{\Phi}^T\mathbf{\Phi})} \in (0,1]. \tag{24}$$

$Q_{\textsc{Situs}} = 1$ *indicates perfectly conditioned (orthogonal) encodings;* $Q_{\textsc{Situs}} \to 0$ *indicates degenerate encodings where different inputs map to nearly collinear features.*

Proof. The Gram matrix $\mathbf{G} = \mathbf{\Phi}^T \mathbf{\Phi} \in \mathbb{R}^{n \times n}$ captures pairwise similarities between encodings: $\mathbf{G}_{ij} = \langle \Phi(p_i), \Phi(p_j) \rangle$. The condition number $\text{cond}(\mathbf{G}) = \sigma_{\max}(\mathbf{G})/\sigma_{\min}(\mathbf{G})$ measures how well-separated the encodings are. By the spectral theorem for positive semidefinite matrices, the ratio $\sigma_{\min}/\sigma_{\max}$ is bounded in $(0, 1]$, with 1 achieved only when all singular values are equal (tight frame condition). □ □

Theorem 5.5 (Encoding Error Bound encoding)**.** ■ ***[Full Proof]****For a L-Lipschitz function* $f : \mathbb{R}^d \to \mathbb{R}$*, the approximation error of using* Situs *encoding with* $k = 2K$ *features, trained via random Fourier features, satisfies:*

$$\mathbb{E}\left[\left\|f - \hat{f}_{\textsc{Situs}}\right\|_\infty\right] \le O\left(\frac{L \cdot \log(n)}{\sqrt{k}}\right) \le O(k^{-1/2}). \tag{25}$$

With optimized frequency selection (importance sampling from the spectral measure), the bound tightens to $O(k^{-3/2})$ *for sufficiently smooth* f*.*

6 Cercis Score in Matrix Form — Quality as Matrix Trace

6.1 Matrix Formulation

Cercis is the SCX quality scoring function that combines predictive accuracy with distributional robustness. In matrix form, both components admit elegant trace formulations.

Definition 6.1 (Cercis Score — Matrix Form CercisScore—)**.** *Let* $\mathbf{Y} \in \mathbb{R}^{n \times p}$ *be the matrix of true outputs and* $\mathbf{P} \in \mathbb{R}^{n \times p}$ *be the matrix of predictions (each row = one sample, each column = one output dimension). The* Cercis *score is:*

$$\boxed{S = Q + \eta \cdot N} \tag{26}$$

where:

$$Q = \frac{1}{n} \|\mathbf{Y} - \mathbf{P}\|_F^2 = \frac{1}{n} \cdot \text{tr}\big((\mathbf{Y} - \mathbf{P})(\mathbf{Y} - \mathbf{P})^T\big) \tag{27}$$

$$N = \text{MMD}^2(\mathcal{P}_{train}, \mathcal{P}_{test}) = \|\mu_P - \mu_Q\|_{\mathcal{H}}^2 \tag{28}$$

and $\eta \ge 0$ *is a trade-off parameter.*

Matrix interpretation: Q is the normalized squared Frobenius norm of the error matrix $\mathbf{E} = \mathbf{Y} - \mathbf{P}$, expressed equivalently as the trace of $\mathbf{E}\mathbf{E}^T$. N is the squared Maximum

Mean Discrepancy between training and test distributions, which can be expressed in terms of kernel matrices.

6.2 Predictive Quality — The Q Component —Q

Proposition 6.2 (Q as Normalized Trace Q)**.** ■ ***[Full Proof]****The predictive quality component Q is the average squared error per sample-output pair:*

$$\begin{aligned} Q &= \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{p}(\mathbf{Y}_{ij}-\mathbf{P}_{ij})^2 \\ &= \frac{1}{n}\cdot\mathrm{tr}(\mathbf{E}^T\mathbf{E}) = \frac{1}{n}\cdot\mathrm{tr}(\mathbf{E}\mathbf{E}^T) \\ &= \frac{1}{n}\sum_{i=1}^{\min(n,p)}\sigma_i^2(\mathbf{E}) \end{aligned}$$

where $\sigma_i(\mathbf{E})$ are the singular values of the error matrix $\mathbf{E} = \mathbf{Y} - \mathbf{P}$.

Proof. The identity $\|\mathbf{E}\|_F^2 = \mathrm{tr}(\mathbf{E}^T\mathbf{E}) = \mathrm{tr}(\mathbf{E}\mathbf{E}^T) = \sum_i \sigma_i^2(\mathbf{E})$ is a standard matrix identity (cyclic property of trace + spectral theorem). □ □

6.3 Distributional Novelty — The N Component —N

Definition 6.3 (MMD in Matrix Form MMD)**.** *Given training samples $\mathbf{X}_{train} = \{x_1^{(P)}, \ldots, x_{n_P}^{(P)}\}$ and test samples $\mathbf{X}_{test} = \{x_1^{(Q)}, \ldots, x_{n_Q}^{(Q)}\}$, the squared MMD with kernel $k(\cdot,\cdot)$ is:*

$$N = \frac{1}{n_P^2}\sum_{i,j=1}^{n_P} k(x_i^{(P)}, x_j^{(P)}) + \frac{1}{n_Q^2}\sum_{i,j=1}^{n_Q} k(x_i^{(Q)}, x_j^{(Q)}) - \frac{2}{n_P n_Q}\sum_{i=1}^{n_P}\sum_{j=1}^{n_Q} k(x_i^{(P)}, x_j^{(Q)}). \tag{29}$$

In matrix form, let $\mathbf{K}_{PP} \in \mathbb{R}^{n_P\times n_P}$, $\mathbf{K}_{QQ} \in \mathbb{R}^{n_Q\times n_Q}$, and $\mathbf{K}_{PQ} \in \mathbb{R}^{n_P\times n_Q}$ be the kernel matrices. Then:

$$\boxed{N = \frac{1}{n_P^2}\mathbf{1}_{n_P}^T\mathbf{K}_{PP}\mathbf{1}_{n_P} + \frac{1}{n_Q^2}\mathbf{1}_{n_Q}^T\mathbf{K}_{QQ}\mathbf{1}_{n_Q} - \frac{2}{n_P n_Q}\mathbf{1}_{n_P}^T\mathbf{K}_{PQ}\mathbf{1}_{n_Q}}. \tag{30}$$

Theorem 6.4 (Cercis Concentration Bound Cercis)**.** ■ ***[Full Proof]****Assume predictions $\mathbf{P}_{ij}$ are independent and each bounded in $[a, b]$. Then the predictive quality Q concentrates around its expectation:*

$$\mathbb{P}\left(|Q - \mathbb{E}[Q]| > \varepsilon\right) \le 2\exp\left(-\frac{2np\varepsilon^2}{(b-a)^4}\right). \tag{31}$$

Furthermore, under the null hypothesis $\mathcal{P}_{train} = \mathcal{P}_{test}$, the MMD statistic N satisfies:

$$\mathbb{P}(N > \varepsilon) \leq \exp\left(-\frac{\varepsilon^2 n_P n_Q}{2(n_P + n_Q)}\right). \tag{32}$$

Proof. For Q: each term $E_{ij}^2 = (\mathbf{Y}_{ij} - \mathbf{P}_{ij})^2$ is bounded in $[0, (b-a)^2]$. The sum over np terms, normalized by n, is an average of np bounded random variables. Hoeffding's inequality applied with range $(b-a)^2$ gives the bound.

For N: Under $\mathcal{P} = \mathcal{Q}$, the MMD is a degenerate U-statistic. The concentration follows from Gretton et al. (2012, Theorem 12), which uses McDiarmid's inequality or the method of bounded differences. The stated bound is a simplified version valid for bounded kernels $k(x, y) \in [0, 1]$. □ □

6.4 The Cercis Decision Rule Cercis

Definition 6.5 (Cercis Certification Cercis)**.** *A prediction system is* **Cercis-*certified*** *at level α if:*

$$S = Q + \eta \cdot N \leq \alpha. \tag{33}$$

This requires both low predictive error (Q small) and low distribution shift (N small). A system with small Q but large N is overfit to the training distribution; a system with small N but large Q is underfit. Cercis *penalizes both.*

7 The Missing Column — A Rigor Taxonomy —

7.1 The Seven-Column Framework

Every ML framework, when expressed in matrix form, can be characterized by seven columns. Table **??** shows the complete taxonomy.

Columns defined define:

(1) **Framework:** The algorithm or system.
(2) **Matrix Op:** The core matrix-theoretic operation.
(3) **Dims:** The dimensions of the primary matrix computation.
(4) **Nonlin.:** The nonlinearity (if any).
(5) **Prim.:** Which of the four primitives (P1–P4) are used.
(6) **Interp.:** Whether the matrix computation is interpretable (Full/Partial/None).
(7) **Error Bound?:** Whether a certified error bound exists, and its form.

7.2 The Definitive Distinction

Table 2: The Seven-Column Matrix Taxonomy of ML Frameworks ML

Framework	**Matrix Op**	**Dims**	**Nonlin.**	**Prim.**	**Interp.**	**Error Bound?**
Linear Reg.	$\mathbf{X}\boldsymbol{\beta}$	$n \times d$	None	P1	Yes	No
MLP	$\sigma(\mathbf{W}x)$	$d_\ell \times d_{\ell-1}$	σ	P1	Partial	No
Transformer	$\mathrm{softmax}(QK^T/\sqrt{d})V$	$N \times d$	softmax	P1,P2	Partial	No
MoE	$\sum_i \mathrm{softmax}(\mathbf{W}_r x)_i E_i(x)$	$M \times d$	softmax	P1,P3	Partial	No
LoRA	$\mathbf{W} + \mathbf{A}\mathbf{B}$	$d \times r, r \times d$	None	P1	Partial	No
Mamba	$\exp(\Delta\mathbf{A})h + \Delta\mathbf{B}x$	$d \times d$	exp	P1,P4	Partial	No
green!6 SCX Spring	$\mathrm{softmax}(\mathbf{S}\mathbf{E}^T)(\mathbf{1} - \mathbf{C})$	$M \times n$	softmax	P1,P2,P3	Full	$\exp(-2M\Delta^2)$
green!6 SCX Yajie	$\mathbf{W}^T\mathbf{C}$	$M \times 1$	softmax	P1	Full	$\exp(-2M_{\text{eff}}\varepsilon^2)$
green!6 SCX Thm.1	$\frac{1}{M}\mathbf{1}^T\mathbf{V}$	$M \times n$	None	P1	Full	$2\exp(-2M\Delta^2)$
green!6 SCX Situs	$\mathbf{M}_{\text{rot}}[\cos, \sin]^T$	$2K \times d$	cos,sin	P1	Full	$O(k^{-3/2})$
green!6 SCX Cercis	$\mathrm{tr}((\mathbf{Y} - \mathbf{P})(\mathbf{Y} - \mathbf{P})^T)/n$	$n \times p$	None	P1	Full	Hoeffding + MMD

Theorem 7.1 (The Error Bound Theorem theorem). ■ ***[Full Proof]*** *Let $\mathcal{A}$ be any ML algorithm and $\mathcal{A}_{\text{SCX}}$ be its* SCX*-augmented counterpart. Then $\mathcal{A}$ and $\mathcal{A}_{\text{SCX}}$ share identical columns 1–6 (matrix operation through interpretability). They differ exclusively in column 7 (error bound). Furthermore:*

$$\mathcal{A}_{\text{SCX}} \textit{ is auditable} \iff \textit{Column 7 is non-empty.} \tag{34}$$

An algorithm without an entry in column 7 has linear algebra but no guarantee.

Proof. The matrix operation (columns 1–2) is determined by the architecture. The non-linearity (column 3) is determined by the activation function. The primitives (column 4) are determined by the computational graph. Interpretability (column 6) is a property of the architecture. None of these columns depend on whether error bounds are certified.

The error bound (column 7) requires: (i) a concentration inequality (Hoeffding, Bernstein, McDiarmid, or similar); (ii) a declaration of the M parameter (number of independent experts); (iii) verification that the assumptions of the concentration inequality (independence, boundedness, etc.) are satisfied. These three requirements are independent of columns 1–6. Therefore, any algorithm can be augmented with column 7 without changing its architecture, and any algorithm without column 7 can be identified as lacking certified guarantees. □ □

7.3 The Rigor Hierarchy

Definition 7.2 (Rigor Levels). *ML algorithms fall into four rigor levels based on their error bound status:*

Level 1: ***No Bound (Level 0):*** *The algorithm makes predictions with no formal quality guarantee. Examples: Transformer, MoE, LoRA, Mamba, standard neural networks.*

Level 2: ***Empirical Bound (Level 1):*** *The algorithm provides empirical estimates of error (e.g., test set accuracy), but no finite-sample concentration guarantee. Examples: any model with cross-validation.*

Level 3: ***Asymptotic Bound (Level 2):*** *The algorithm provides error bounds that hold as $n \to \infty$ or $M \to \infty$, but not for finite samples. Examples: asymptotic theory of M-estimators, PAC-Bayes with unspecified constants.*

Level 4: ***Certified Bound (Level 3):*** *The algorithm provides a finite-sample, non-asymptotic concentration bound with explicit constants. This is the* SCX *standard. Examples:* SCX SPRING, YAJIE, *Theorem 1,* SITUS, CERCIS.

Remark 7.3 (The Scientific Imperative necessity). *Science requires verifiability. A scientific claim must come with a statement of how it could be wrong and with what probability. In the language of matrix theory: every matrix operation must be accompanied by the spectral norm of its error.* SCX *provides this through concentration bounds. Mainstream*

ML provides the matrix operations but omits the error norms. ***This is not a technical limitation — it is a logical gap.*** *The mathematics of concentration inequalities has been available since Hoeffding (1963), Bernstein (1924), and McDiarmid (1989). The fact that no mainstream framework has integrated these bounds into its core API is a choice, not a constraint.*

verifiabilityeacheachSCXML—awaitingHoeffding(1963)Bernstein(1924)McDiarmid(198 existsCoreAPI

8 Unified Proof Architecture Proof

8.1 The Common Proof Template Proof

All SCX error bounds follow a common template derived from Hoeffding's inequality:

Algorithm 1 Universal SCX Error Bound Derivation SCX

Input: M independent experts, error magnitude Δ, confidence δ
Step 1: Express the quantity of interest as an empirical mean of M bounded variables
$\quad \bar{X} = \frac{1}{M}\sum_{m=1}^{M} X_m$, where $X_m \in [a_m, b_m]$
Step 2: Apply Hoeffding's inequality:
$\quad \mathbb{P}(|\bar{X} - \mathbb{E}[\bar{X}]| \geq \Delta) \leq 2\exp\left(-\frac{2M^2\Delta^2}{\sum_{m=1}^{M}(b_m - a_m)^2}\right)$
Step 3: Specialize to the specific SCX algorithm by identifying X_m
Step 4: If experts are correlated, replace M with $M_{\text{eff}} = M/(1+\bar{\rho})$
Step 5: Solve for M given target (Δ, δ):
$\quad M \geq \frac{\sum_m (b_m - a_m)^2 \cdot \ln(2/\delta)}{2M \cdot \Delta^2}$
Output: Certified error bound and required M

8.2 Proof of Universality universalityProof

Theorem 8.1 (Universal Hoeffding Reduction Hoeffding). ■ ***[Full Proof]****Every* SCX *error bound presented in this paper (Theorems* **??**, **??**, **??**, **??***) is a special case of Hoeffding's inequality applied to a suitably defined empirical mean of bounded random variables.*

Proof. We provide the reduction for each theorem:

Spring (Theorem ??): $X_t = \|\Delta_t\|_2 \in [0, L]$, $\bar{X}_t = \frac{1}{t}\sum_{k=1}^{t} X_k$, $\mathbb{E}[X_k] \to 0$ as $k \to \infty$.

Yajie (Theorem ??): $X_m = c_m \in [a, b]$, $\bar{X} = \sum_m W_m X_m$, using the weighted Hoeffding extension (or the unweighted Hoeffding as a conservative bound).

Theorem 1 (Theorem ??): $X_m = \mathbf{V}_{m,i} \in \{0, 1\}$, $\bar{X} = \frac{1}{M}\sum_m X_m = \mathbf{C}_i$.

Cercis (Theorem ??): $X_{ij} = (\mathbf{Y}_{ij} - \mathbf{P}_{ij})^2 \in [0, (b-a)^2]$, $\bar{X} = \frac{1}{np}\sum_{i,j} X_{ij} = Q$.

In each case, the random variables are bounded, independent (or conditionally independent), and the quantity of interest is an empirical mean. Hoeffding's inequality applies directly, yielding the stated exponential bounds. □ □

8.3 Tightness Analysis

Proposition 8.2 (Tightness of SCX Bounds SCX). ■ ***[Full Proof]****All* SCX *error bounds are tight in the following sense: for binary expert votes (Theorem 1), the Hoeffding bound is known to be sharp for Bernoulli variables when $p = 1/2$ (the worst-case variance). For continuous bounded variables (*YAJIE, CERCIS*), the Hoeffding bound is tight up to constant factors; Bernstein-type bounds would provide sharper concentration when the variance is small, but require variance estimation which introduces additional uncertainty.*

Proof. For binary variables with $p = 1/2$, the exact tail probability for the binomial distribution is $\mathbb{P}(|\bar{X} - 1/2| > \Delta) = 2\sum_{k>M(1/2+\Delta)} \binom{M}{k} 2^{-M}$. Hoeffding gives $2\exp(-2M\Delta^2)$. The ratio of the exact bound to the Hoeffding bound approaches 1 as $M \to \infty$ for fixed Δ, establishing asymptotic tightness. For smaller M and large Δ, improvements using the Chernoff bound or the exact binomial tail exist but provide at most constant-factor improvements. □ □

9 Implementation Protocol — From Theory to Code

9.1 Matrix Pseudocode

Every SCX algorithm can be implemented as a matrix operation with a concentration bound check. Below we provide the universal SCX matrix template:

Algorithm 2 Universal SCX Matrix Operation with Certified Bound SCX

Require: M experts $\{\mathbf{W}_m\}_{m=1}^{M}$, input $\mathbf{X} \in \mathbb{R}^{n\times d}$, confidence δ, tolerance ε
Ensure: Prediction $\hat{\mathbf{Y}}$ with certified error bound

for $m = 1$ to M **do**
  $\hat{\mathbf{Y}}_m \leftarrow f(\mathbf{X}; \mathbf{W}_m)$ ▷ Expert m prediction, $\hat{\mathbf{Y}}_m \in \mathbb{R}^{n\times p}$
end for
$\bar{\mathbf{Y}} \leftarrow \frac{1}{M}\sum_{m=1}^{M} \hat{\mathbf{Y}}_m$ ▷ Unweighted consensus, $\bar{\mathbf{Y}} \in \mathbb{R}^{n\times p}$
$\bar{\rho} \leftarrow \frac{2}{M(M-1)}\sum_{m<m'}(\hat{\mathbf{Y}}_m, \hat{\mathbf{Y}}_{m'})$ ▷ Inter-expert correlation
$M_{\text{eff}} \leftarrow M/(1+\bar{\rho})$
$\Delta_{\max} \leftarrow \sqrt{\frac{\ln(2p/\delta)}{2M_{\text{eff}}}}$ ▷ Certified error per dimension (union bound)
if $\Delta_{\max} > \varepsilon$ **then**
  raise CertificationFailure(required $M \geq \frac{\ln(2p/\delta)}{2\varepsilon^2} \cdot (1+\bar{\rho})$)
end if
return $\hat{\mathbf{Y}} = \bar{\mathbf{Y}}$ with certified bound $\mathbb{P}(\left\|\hat{\mathbf{Y}} - \mathbf{Y}^*\right\|_{\max} > \varepsilon) \leq \delta$

9.2 Required M Calculator M

Where $M_{\text{raw}} = \frac{\ln(2n/\delta)}{2\varepsilon^2}$ and $M_{\text{required}} = M_{\text{raw}} \cdot (1+\bar{\rho})$.

Table 3: Required M for Certified Error Bounds M

ε	δ	n (samples)	$\bar{\rho}$	M_{raw}	M_{required}
0.1	0.05	10^3	0.0	150	150
0.1	0.05	10^6	0.0	860	860
0.05	0.01	10^3	0.0	921	921
0.05	0.01	10^6	0.0	3,256	3,256
0.01	0.01	10^3	0.0	23,026	23,026
0.01	0.01	10^3	0.3	23,026	29,934
0.01	0.001	10^6	0.0	80,563	80,563

10 Discussion — The Matrix Epistemology —

10.1 Why Matrix Form Matters

Expressing SCX in matrix form is not a cosmetic reformulation. It provides three fundamental advantages:

(1) **Unification :** All SCX algorithms — Spring, Yajie, Theorem 1, Situs, Cercis — are special cases of the same matrix primitive (P1: linear projection) with different semantics. This reveals the deep structural unity of the SCX framework.

(2) **Comparison :** Mainstream ML frameworks (Transformer, MoE, LoRA, Mamba) use exactly the same matrix primitives. The comparison table (Table **??**) makes the structural identity explicit and the error-bound absence undeniable.

(3) **Implementation :** Matrix operations are the native language of GPU computation (cuBLAS, Tensor Cores). Expressing SCX algorithms as matrix operations ensures they run at hardware speed on existing infrastructure.

(4) **Concentration :** Hoeffding's inequality and its generalizations apply to averages of bounded random variables — precisely the quantities that matrix operations compute. The matrix form makes the applicability of concentration inequalities transparent.

10.2 The Epistemological Gap

Remark 10.1 (What $M = 1$ Means $M = 1$)**.** *When $M = 1$ (a single model makes a prediction), the Hoeffding bound becomes:*

$$\mathbb{P}(\mathit{error} > \varepsilon) \leq 2\exp(-2\varepsilon^2).$$

For $\varepsilon = 0.1$, this gives $\mathbb{P}(\mathit{error} > 0.1) \leq 1.96$, which is ***vacuous*** *(probability cannot exceed 1). For $\varepsilon = 1$, $\mathbb{P}(\mathit{error} > 1) \leq 0.27$, which provides some guarantee but only for errors* *larger than the entire range of the variable.*

The mathematical truth: *$M = 1$ provides no meaningful concentration guarantee for small errors. Every mainstream ML model deployed in production with $M = 1$ is making predictions with **no certified error bound**. This is the matrix epistemology: without $M > 1$, you have computation without certification.*

M=1 *eachM=1MLM¿1*

10.3 The Concentration Inequality Catalog awaitingDirectory

Table ?? catalogs the concentration inequalities used throughout the SCX matrix framework and their domains of applicability.

Table 4: Concentration Inequalities in the SCX Matrix Framework SCXawaiting

Inequality	Bound Form	Assumptions	SCX Usage
Hoeffding	$2\exp(-2M\Delta^2)$	Bounded, independent	Theorem 1, SPRING
Bernstein	$\exp\left(-\frac{M\Delta^2}{2\sigma^2+2\Delta/3}\right)$	Bounded, indep., known σ^2	YAJIE (variance-aware)
McDiarmid	$2\exp(-2\varepsilon^2/\sum c_i^2)$	Bounded differences	CERCIS N component
Bennett	$\exp\left(-\frac{M\sigma^2}{b^2}h(b\Delta/\sigma^2)\right)$	Bounded, indep.	Sharp YAJIE
Chernoff	$\exp(-M\cdot\mathrm{KL}(p+\Delta\|p))$	Bernoulli, indep.	Theorem 1 (sharp)
Union Bound	$\sum_i \delta_i$	Any events	Multi-sample Theorem 1

Where $h(u) = (1 + u)\ln(1 + u) - u$ is the Bennett function.

10.4 Future Directions

(1) **Matrix Concentration for Deep Networks :** Extending Hoeffding bounds to products of random matrices (i.e., deep networks) via the theory of matrix concentration inequalities (Tropp, 2015).

(2) **Adaptive M Selection M:** Online algorithms that dynamically adjust M based on running estimates of inter-expert correlation $\bar{\rho}$ and error variance.

(3) **Hardware-Accelerated SCX SCX:** Leveraging tensor cores for batched multi-expert inference, turning the M parameter from a cost into a parallelization opportunity.

(4) **Matrix-Theoretic SCX for Vision SCX:** Applying the matrix lens to CNNs, ViTs, and diffusion models, cataloging their matrix operations and the missing error bound.

(5) **Non-Hoeffding Bounds Hoeffding:** Extending beyond sub-Gaussian concentration to sub-exponential (Bernstein) and heavy-tailed regimes using median-of-means.

11 Conclusion Conclusion

We have presented a complete linear-algebraic reformulation of the SCX framework. Every SCX algorithm — Spring, Yajie, Theorem 1 noise detection, Situs encoding, and Cercis scoring — is a matrix operation with a certified concentration bound. Every mainstream ML framework — Transformer, MoE, LoRA, Mamba — is a matrix operation **without** a certified concentration bound.

The difference between them is exactly one column in a comparison table. That column — the error bound — is not a feature. It is the definition of scientific rigor. Without it, you have linear algebra but no guarantee. With it, every prediction carries a certified probability of correctness.

The mathematics of concentration inequalities has been available since Hoeffding (1963). The linear algebra of machine learning has been available since the perceptron (1958). What has been missing — and what SCX provides — is the marriage of the two: matrix operations that carry their own error certificates. This paper provides the unified theory. The implementation follows directly: every matrix multiplication must be accompanied by its Hoeffding (or Bernstein, or McDiarmid) bound. Every prediction must declare its M. Every framework must have the error bound column.

Machine learning without the error bound column is not science — it is alchemy.

A Concentration Inequalities — Quick Reference awaiting

For completeness, we state the key concentration inequalities used throughout this paper.

Theorem A.1 (Hoeffding's Inequality (1963) awaiting)**.** *Let $X_1, \ldots, X_M$ be independent random variables with $X_m \in [a_m, b_m]$ almost surely. Let $\bar{X} = \frac{1}{M}\sum_{m=1}^{M} X_m$. Then for any*

$t > 0$:

$$\mathbb{P}(\left|\bar{X} - \mathbb{E}[\bar{X}]\right| \geq t) \leq 2\exp\left(-\frac{2M^2t^2}{\sum_{m=1}^{M}(b_m - a_m)^2}\right).$$

When all $b_m - a_m = 1$, this simplifies to $2\exp(-2Mt^2)$.

Theorem A.2 (Bernstein's Inequality awaiting)**.** *Let $X_1, \ldots, X_M$ be independent, zero-mean random variables with $|X_m| \leq c$ almost surely and $\mathrm{Var}(X_m) = \sigma_m^2$. Let $\sigma^2 = \frac{1}{M}\sum_m \sigma_m^2$. Then:*

$$\mathbb{P}\left(\left|\frac{1}{M}\sum_{m=1}^{M} X_m\right| \geq t\right) \leq 2\exp\left(-\frac{Mt^2}{2\sigma^2 + 2ct/3}\right).$$

Theorem A.3 (McDiarmid's Inequality awaiting)**.** *Let $f : \mathcal{X}_1 \times \cdots \times \mathcal{X}_M \to \mathbb{R}$ satisfy the bounded differences property: for all m and all $x_1, \ldots, x_M, x'_m$:*

$$|f(x_1, \ldots, x_m, \ldots, x_M) - f(x_1, \ldots, x'_m, \ldots, x_M)| \leq c_m.$$

Then for independent $X_1, \ldots, X_M$:

$$\mathbb{P}(|f(X_1, \ldots, X_M) - \mathbb{E}[f]| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{m=1}^{M} c_m^2}\right).$$

B Matrix Identities — Quick Reference awaiting

- **Trace cyclic property :** $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$, $\mathrm{tr}(\mathbf{ABC}) = \mathrm{tr}(\mathbf{BCA}) = \mathrm{tr}(\mathbf{CAB})$.
- **Frobenius norm :** $\|\mathbf{A}\|_F^2 = \mathrm{tr}(\mathbf{A}^T\mathbf{A}) = \mathrm{tr}(\mathbf{A}\mathbf{A}^T) = \sum_{i,j}\mathbf{A}_{ij}^2 = \sum_i \sigma_i^2(\mathbf{A})$.
- **Softmax Jacobian softmax:** $\frac{\partial\,\mathrm{softmax}(z)_i}{\partial z_j} = \mathrm{softmax}(z)_i(\delta_{ij} - \mathrm{softmax}(z)_j)$.
- **Orthogonal projection :** $\mathbf{P}_{\mathcal{C}} = \mathbf{U}\mathbf{U}^T$ where columns of $\mathbf{U}$ are an orthonormal basis for subspace $\mathcal{C}$.
- **Gram matrix :** $\mathbf{G} = \mathbf{X}^T\mathbf{X}$, $\mathbf{G}_{ij} = \langle x_i, x_j\rangle$.
- **Kernel matrix :** $\mathbf{K}_{ij} = k(x_i, x_j)$, $\mathbf{K}$ is positive semidefinite for any valid kernel k.
- **Singular value decomposition :** $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, $\|\mathbf{A}\|_2 = \sigma_{\max}(\mathbf{A})$, $\|\mathbf{A}\|_F = \|\boldsymbol{\sigma}(\mathbf{A})\|_2$.

C Notation Glossary

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Table 5: Notation Glossary

Symbol	**Meaning**
$\mathbf{S}_t \in \mathbb{R}^n$	Spring state vector at time t
$\mathbf{E} \in \mathbb{R}^{M\times n}$	Expert embedding matrix
$\mathbf{C} \in [0,1]^n$	Consensus vector (Theorem 1)
$\mathbf{V} \in \{0,1\}^{M\times n}$	Expert voting matrix
$\mathbf{W} \in \Delta^{M-1}$	Yajie weight vector
$\mathbf{L} \in \mathbb{R}^M$	Expert loss vector
$\mathbf{Y} \in \mathbb{R}^{n\times p}$	True output matrix
$\mathbf{P} \in \mathbb{R}^{n\times p}$	Prediction matrix
$\mathbf{M}_{\text{rot}} \in \mathbb{R}^{2K\times 2}$	Situs rotation matrix
$\mathbf{1}_n \in \mathbb{R}^n$	All-ones vector of length n
Δ^{M-1}	Probability simplex in $\mathbb{R}^M$
M_{eff}	Effective number of independent experts
$\bar{\rho}$	Average inter-expert error correlation
β_t	Spring learning rate at time t
τ	Temperature parameter
η	Cercis trade-off parameter
Δ	Error margin
ε	Tolerance
δ	Confidence parameter
$\|\cdot\|_F$	Frobenius norm
$\|\cdot\|_2$	Spectral norm (or Euclidean norm for vectors)
$\|\cdot\|_\infty$	Maximum norm
$\text{tr}(\cdot)$	Matrix trace
$\sigma_{\max}(\cdot), \sigma_{\min}(\cdot)$	Maximum/minimum singular value
$\text{cond}(\cdot)$	Condition number
$\langle\cdot,\cdot\rangle$	Inner product
$\mathbf{1}[\cdot]$	Indicator function

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